

Bluestock Mutual Fund Analytics Capstone Project

- Submitted By: Sujal Patel
 - Project Type: Data Analytics & Business Intelligence
 - Technologies: Python, SQLite, Power BI, Pandas, Matplotlib
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Executive Summary

This page explains:

- Project objective
 - Data engineering workflow
 - Analytics performed
 - Dashboard implementation
 - Business value generated
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Introduction & Problem Statement

Introduction

The mutual fund industry generates large volumes of financial and investor data.

Analyzing this information manually is difficult and time-consuming.

Problem Statement

The project aims to:

- Analyze mutual fund performance
- Understand investor behavior
- Monitor industry growth
- Evaluate investment risk
- Build interactive dashboards

Objectives

1. Create ETL pipeline
2. Build data warehouse
3. Perform EDA
4. Create dashboards
5. Perform advanced analytics
6. Develop fund recommender system

Data Sources

Datasets Used

Dataset	Purpose
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dim_fund	Fund details
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dim_date	Calendar data
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fact_nav	Daily NAV history
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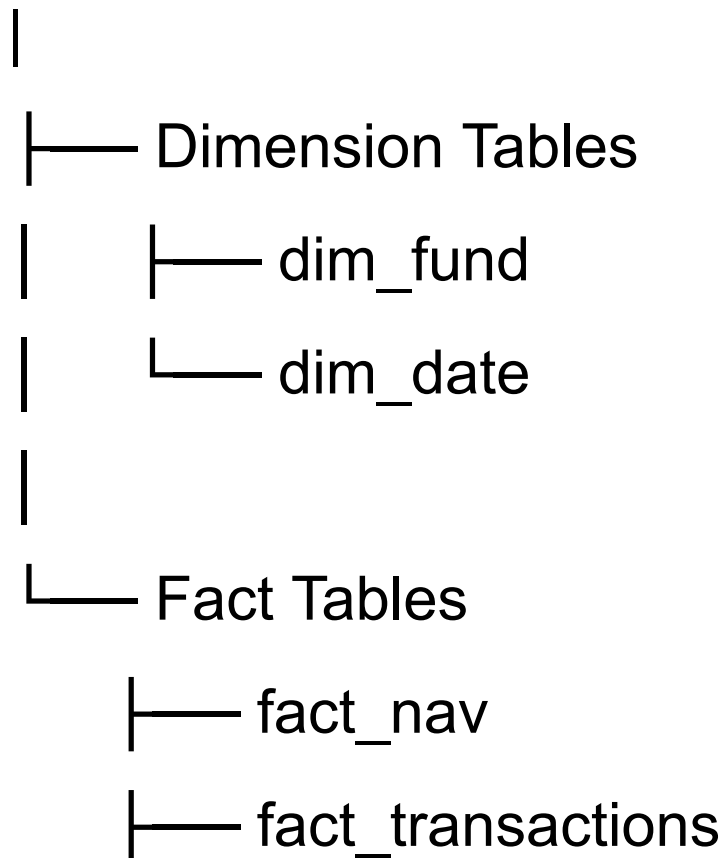
fact_transactions	Investor transactions
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fact_per Fund
formanc performa
e nce

fact_au Industry
m AUM

Database Design

SQLite Data Warehouse



└── fact_performance
└── fact_aum

ETL Design

ETL Workflow

Raw CSV Files



Extract



Transform



Validation



SQLite Warehouse



Dashboard & Analytics

Key Transformations

- Missing value handling
- Duplicate removal
- Data type conversion
- Standardization
- Validation checks

ETL Benefits

- Automation
- Consistency
- Data quality
- Scalability

Exploratory Data Analysis (EDA)

Key Findings

Industry Growth

- Industry AUM exceeded ₹81 Lakh Crore.
- Continuous growth observed.

Leading AMCs

1. SBI Mutual Fund
2. ICICI Prudential MF
3. HDFC MF
4. Nippon India MF
5. Kotak MF

Fund Performance

- Majority of funds delivered positive returns.
 - Equity funds outperformed debt funds.
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Investor Analytics

State-wise Analysis

Top contributors:

- Maharashtra

- Delhi
- Karnataka
- Gujarat
- Tamil Nadu

Age Group Analysis

Age Group	Observation
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18–25	Low investment
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26–35	Highest SIP activity
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36–4 Highest
5 investmen
t

46–6 Stable
0 investors

60+ Conservati
ve
investors

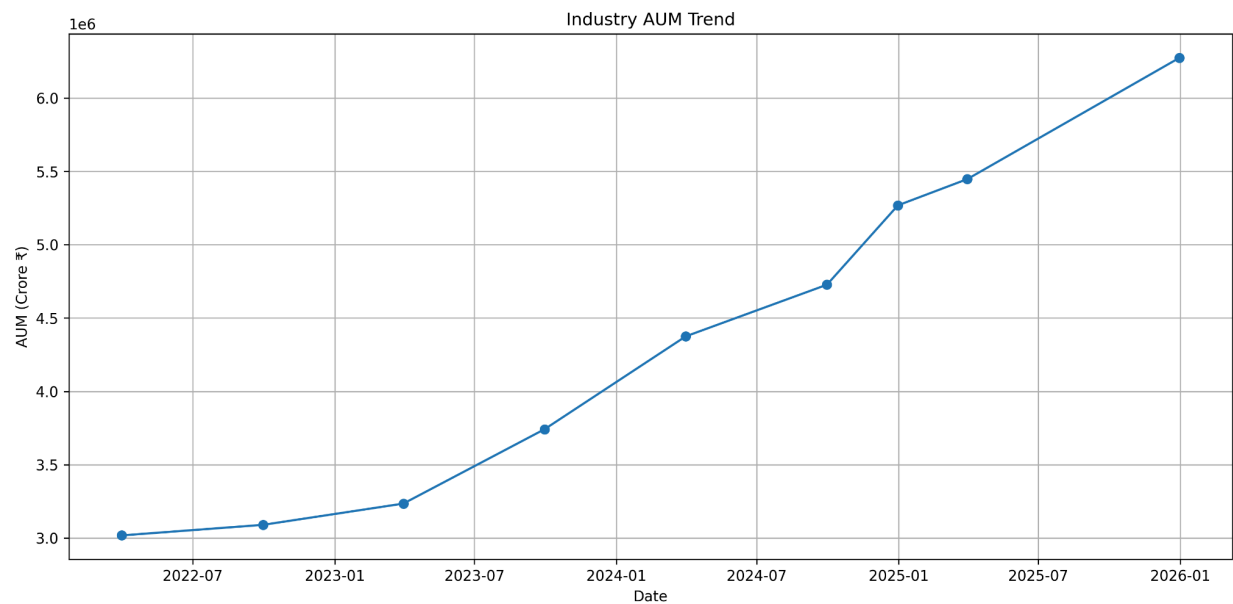
Transaction Analysis

- SIP dominates transactions.
- Lumpsum contributes larger ticket sizes.
- Redemptions are relatively lower.

Dashboard Design

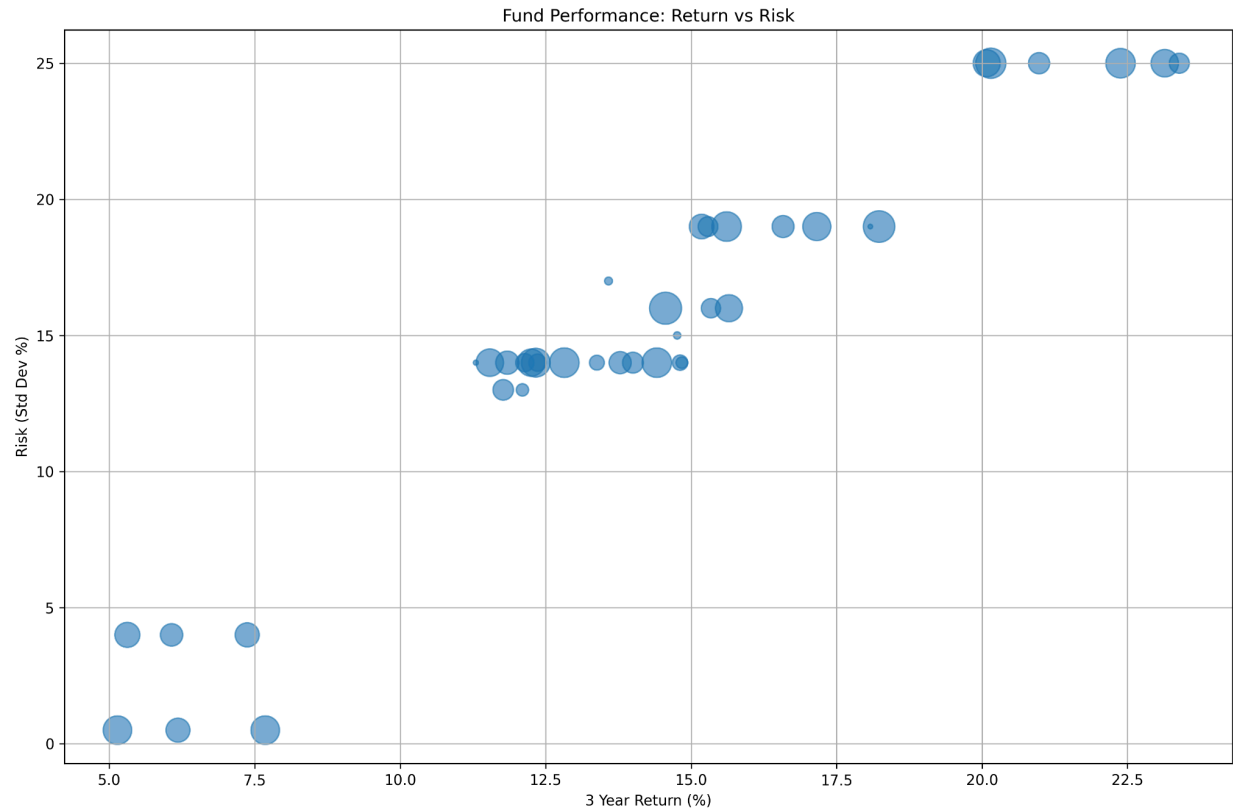
Dashboard Pages

Industry Overview



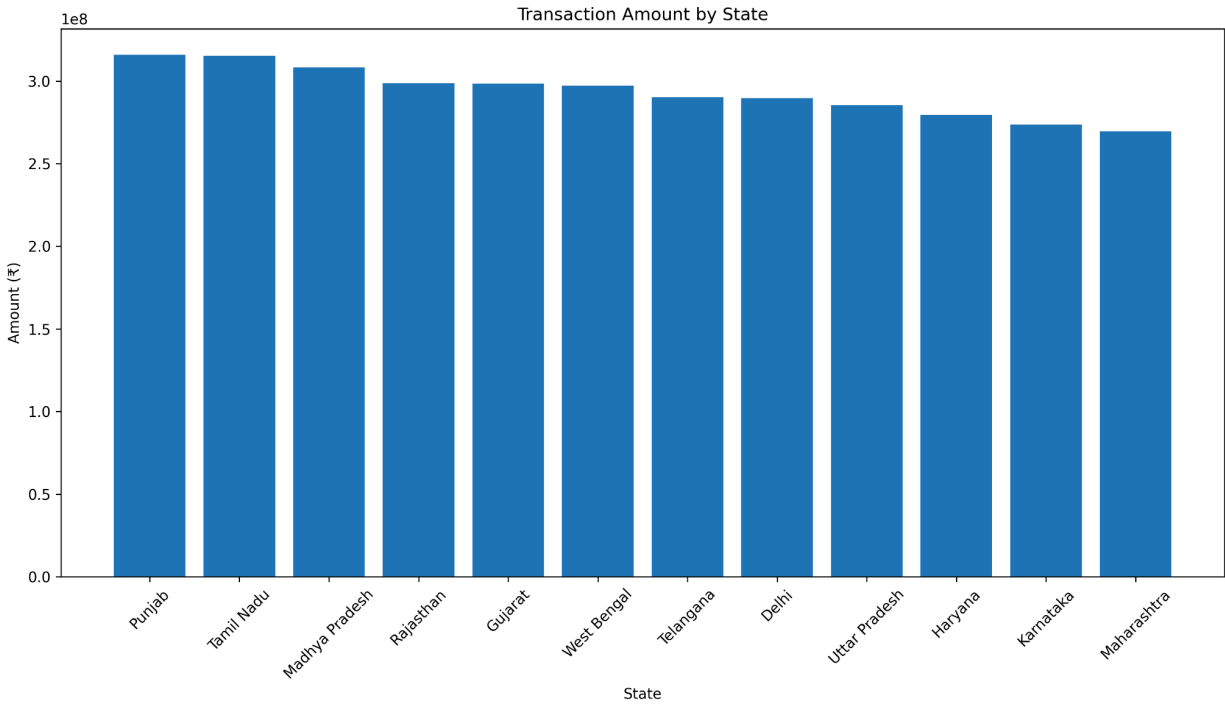
- KPI Cards
- AUM Trend
- AUM by AMC

Fund Performance



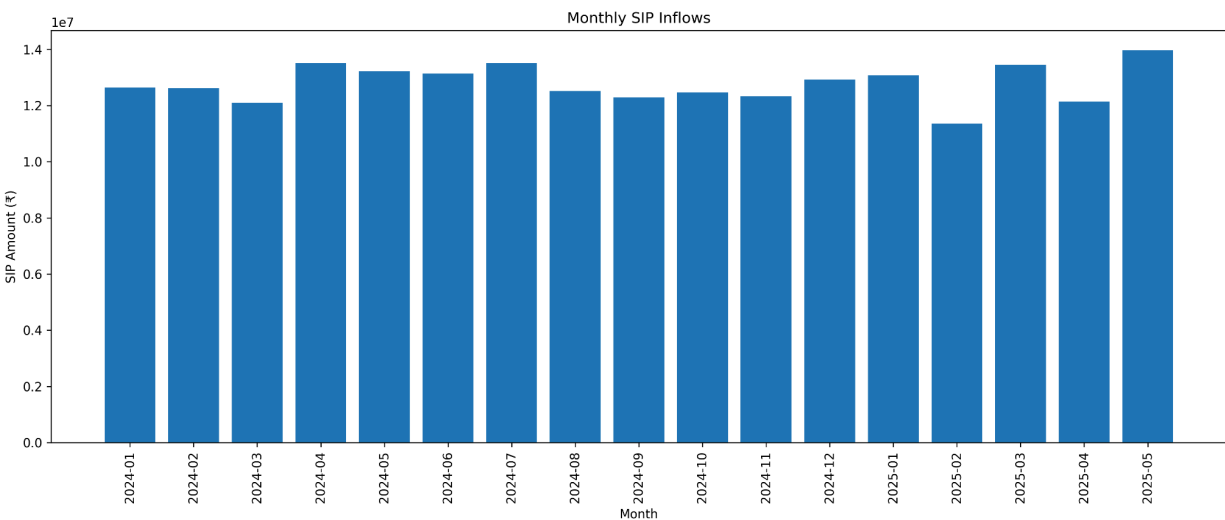
- Risk vs Return Scatter Plot
- NAV Trends
- Fund Scorecard

Investor Analytics



- State-wise Investments
- SIP/Lumpsum Distribution
- Monthly Transactions

SIP & Market Trends



- SIP vs Nifty Trend
 - Category Inflows
 - Top Categories
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Advanced Analytics

Historical VaR & CVaR

Measures downside risk.

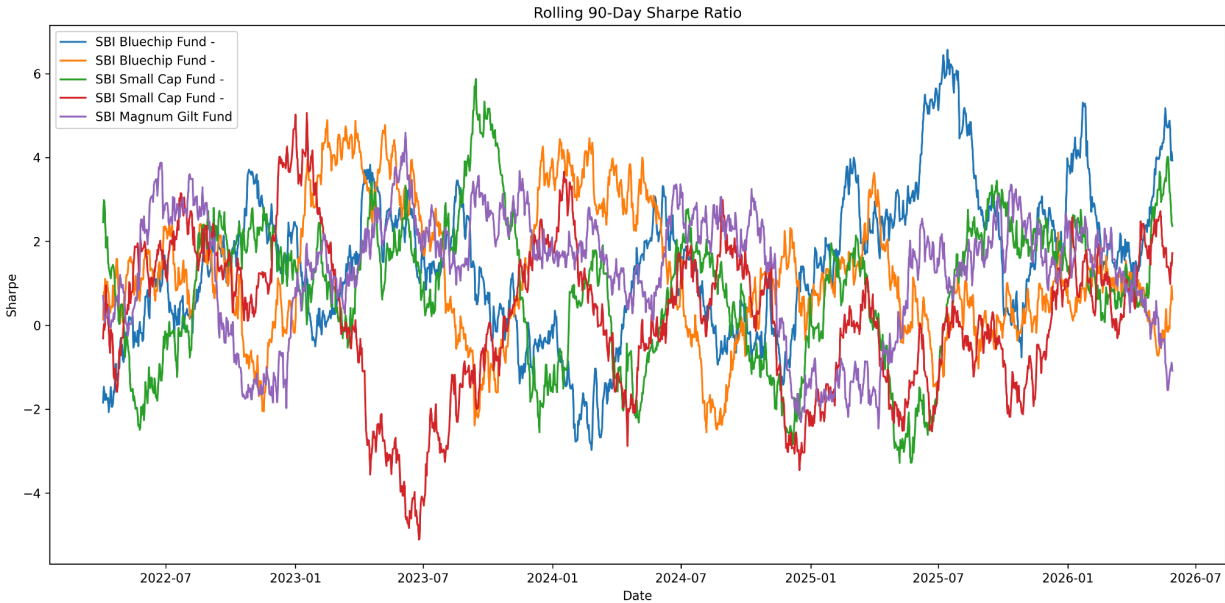
Formula

$VaR_{95} = 5\text{th Percentile}(\text{Return Distribution})$

Benefits

- Risk estimation
 - Capital preservation
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Rolling Sharpe Ratio



Sharpe Ratio = (Rolling Mean Return ÷
Rolling Standard Deviation) × $\sqrt{252}$

Measures risk-adjusted performance over time.

Investor Cohort & SIP Continuity Analysis

Cohort Analysis

Investors grouped by first transaction year.

Metrics:

- Average SIP Amount

- Total Invested
- Preferred Fund

SIP Continuity

Investor classified as:

Active

Average SIP gap $\leq$ 35 days

At Risk

Average SIP gap $>$ 35 days

Benefits:

- Churn prediction
- Investor retention

Fund Recommendation System & Key Findings

Fund Recommendation Logic

Input:

Low Risk

Moderate Risk

High Risk

Output:

Top 3 funds ranked by Sharpe Ratio.

Key Findings

1. Industry AUM growing steadily.
 2. SIP is the dominant investment mode.
 3. Maharashtra contributes highest investments.
 4. Investors aged 26–45 are most active.
 5. Equity funds remain the preferred category.
 6. Risk-adjusted metrics provide better fund selection.
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Conclusion & Recommendations

Conclusion

The Bluestock Mutual Fund Analytics Platform successfully combines:

- Data Engineering
- Data Warehousing
- Financial Analytics
- Business Intelligence

to provide actionable investment insights.

Recommendations

Short-Term

- Improve dashboard automation
- Add more investor segmentation

Long-Term

- Live AMFI integration
- Real-time NAV updates
- Machine Learning fund recommender
- Investor churn prediction

Future Scope

- Cloud deployment
- Power BI Service publishing
- Predictive analytics
- Portfolio optimization

Thank You